Retention-constrained class expansion with reference-anchored candidate tables

Junyi Zhang[a,*]

[a]*Shandong University, China*

Abstract

Reference-anchored candidate tables reuse a trained classifier to recognize new classes under an original-class accuracy requirement. Fixed base references preserve existing scores; query-dependent normalization combines learned, semantic and prototype evidence, and a separate threshold controls novel admission. Controlled normalization experiments isolate the value of query scaling. Across three datasets and two frozen CLIP encoders, matched-retention frontiers and paired arrivals quantify classifier value. Independent calibration compares protection loss with a direct accuracy test that credits recovered errors. On ImageNet-R ViT-L/14, the direct test admits both retained and dense-prototype readouts in all ten partitions; novel accuracy is 81.61% versus 76.76%. Query-shift experiments establish the role of calibration–target correspondence.

Keywords: Class expansion, Retention constraints, Incremental learning, Evidence fusion, Vision–language models, Long-tailed recognition

*Corresponding author.

Email address: cheerssom0876@gmail.com (Junyi Zhang)

1. Introduction

An expanding recognition system already has a classifier whose decision boundaries reflect task-specific supervision. New classes arrive with names and a few examples. The central question is how much new-class recognition that existing discrimination can support while maintaining a specified accuracy on the original classes. A second question follows across repeated arrivals: how much predictive value is gained by keeping the trained classifier rather than replacing it with prototypes or recalibrated semantic evidence?

We address these questions with *reference-anchored candidate tables*. The table retains learned base columns and appends semantic anchors and support prototypes for arriving classes. Each evidence branch is normalized against fixed base references, so adding candidates leaves existing scores unchanged. An independent admission offset controls competition between base and novel classes. This separates the discrimination already learned from the decision to admit new alternatives.

The contribution is the connection between retained discrimination and class-expansion capacity: a persistent score frame makes the original classifier reusable, a common base-accuracy constraint makes its admission value comparable, and paired arrival sequences measure its continuing benefit. Independent calibration tests the shared accuracy requirement through either a sufficient protection loss or a direct paired accuracy test that credits recovered errors. A controlled centering-by-scaling experiment isolates the query-dependent normalization contribution. On ImageNet-R, the gain over dense prototypes persists across two frozen encoders. With sufficient independent calibration coverage, the tables certify final-session expansion on every

evaluated partition of all three datasets and both encoders.

2. Related work

Long-tailed recognition addresses unequal class supervision through balanced fitting and losses [1, 2, 3]. CLIP supplies semantic image–text evidence [4], while prompt adaptation [5, 6], prototypes [7, 8] and semantic caches [9] adapt pre-trained representations to labeled data. Our setting combines these evidence types according to their availability at class arrival.

Class-incremental and few-shot incremental learning study retention across successive tasks [10, 11]. SimpleCIL [12] and RanPAC [13] show the value of pre-trained prototypes and analytic readouts. KBS approaches develop multi-feature similarity supplementation [14], sharpness-aware prototype calibration [15], adaptive exploration [16], representation-enhanced analytic learning [17], and multimodal prompt fusion [18].

Recent work aligns task-specific expert inference in MoTE [19], separates expanding feature spaces through orthogonal constraints [20], and learns invariant representations through counterfactual distribution intervention [21]. Our contribution concerns the admission capacity supplied by an existing classifier under a common base-accuracy requirement. CLIP-guided discovery [22] additionally uses unlabeled novel data, whereas our arriving classes provide names and labeled support. FeNeC [23] supplies a clustered-representative readout for controlled frozen-feature comparisons.

BiMC [24] and HyCal [25] calibrate multimodal or heterogeneous recognition metrics. Calibrated seen/unseen recognition [26] studies the related operating trade-off, and Learn then Test [27] supplies risk-controlled thresh-

old selection. We focus on the retained classifier as a source of admission capacity: its contribution is measured at a common base requirement, over successive arrivals, and under separately calibrated protection and direct accuracy policies. The comparison includes an author-code-checked HyCal transfer with matched encoders, labels and support.

3. From retained discrimination to novel admission

3.1. Task and evidence

Base classes $\mathcal{B}$ have labeled, potentially imbalanced fitting data. Arriving classes $\mathcal{N}$ provide a name and k labeled support images per class. Prediction selects from $\mathcal{B} \cup \mathcal{N}$ without a group oracle. A frozen encoder produces a normalized feature $v = f(x)$; normalized text anchors t_c average eight fixed prompts. Learned base scores are $q_L(x, b) = v^\top w_b + b_b$, fitted by inverse-frequency-weighted ridge regression. Novel prototypes are normalized support means p_n, giving $q_P(x, n) = v^\top p_n$; semantic scores are $q_T(x, c) = v^\top t_c$.

For each evidence source e, a fixed base reference bank $\mathcal{R}_e$ defines

$$\mu_e(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} q_e(x, b), \tag{1}$$

$$\sigma_e^2(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} (q_e(x, b) - \mu_e(x))^2. \tag{2}$$

We use base text anchors, learned columns and visual means as the text, learned and prototype references. Normalized evidence is

$$\widehat{q}_e(x, c) = \frac{q_e(x, c) - \mu_e(x)}{\max(\sigma_e(x), 10^{-6})}. \tag{3}$$

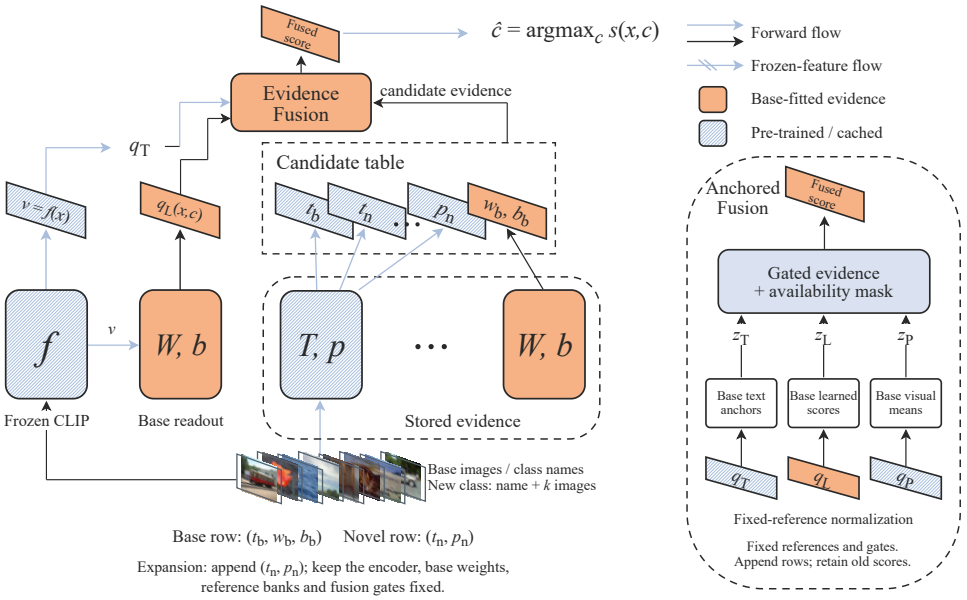


Figure 1: Reference-anchored candidate tables. Fixed base references set each branch's score frame; learned base columns remain stored as novel text anchors and support prototypes are appended. The separate novel offset controls admission. Orange marks base-fitted evidence; blue hatching marks pre-trained or cached evidence.

The statistics adapt to each query while remaining independent of the active novel vocabulary. Fusion uses the evidence available to each row:

$$s(x,b) = \widehat{q}_T(x,b) + a\widehat{q}_L(x,b), \qquad b \in \mathcal{B}, \tag{4}$$

$$s(x,n) = \widehat{q}_T(x,n) + u\widehat{q}_P(x,n) + d, \qquad n \in \mathcal{N}. \tag{5}$$

The coefficients a, u combine evidence; d controls novel admission (Fig. 1). Candidate-wise normalization instead changes the branch scales when new rows arrive, which can alter existing fused decisions. A numerical example appears in Supplementary Section K.

Proposition 1. *With fixed input, encoder, existing row evidence, reference*

banks, coefficients and tie-breaking, appending rows preserves every existing score and its within-vocabulary ordering.

All terms of an existing row remain fixed. This structural property also holds for raw and fixed-temperature controls; their recognition comparisons isolate reference normalization. Appended rows can still win the expanded prediction. Supplementary Section E gives the proof and prediction consequence. Each new row costs $O(D)$ storage and $O(kD)$ prototype construction after encoding; scoring K classes costs $O(KD)$.

3.2. Calibration and expansion capacity

A base-only pseudo-expansion episode splits base classes into pseudo-base and pseudo-novel groups. A classifier fitted on pseudo-base classes is combined with names and reserved support for pseudo-novel classes. Held-out base queries select a, u, d by harmonic macro accuracy,

$$H = 2A_B A_N/(A_B + A_N), \tag{6}$$

with zero assigned when both accuracies vanish. The coefficients then remain fixed across actual arrivals. Actual novel query labels and test labels are excluded from this selection; grids and split seeds are in Supplementary Section K.

Let $A_{B,\mathrm{ref}}^{\mathrm{closed}}$ denote the retained table's closed-base macro accuracy. At retention fraction r, every readout receives the same absolute base target. Its empirical expansion capacity is

$$N_r = \max_d A_N(d) \quad \text{subject to} \quad A_B(d) \geq r A_{B,\mathrm{ref}}^{\mathrm{closed}}. \tag{7}$$

A test-label sweep characterizes this descriptive frontier; an operational threshold uses calibration data. The exact margin envelope is derived in Supplementary Section D. Reporting these two evaluations separately distinguishes attainable capacity from selected-policy behavior.

3.3. Independent accuracy calibration

Let R indicate correct prediction by the retained closed-base reference and C_d correct prediction by the expanded readout. Define

$$L_d = R(1 - C_d) + (1 - r)(1 - R). \tag{8}$$

Writing $D = \Pr(R = 1, C_d = 0)$ gives $\mathbb{E}L_d = D + (1 - r)(1 - A_{B,\mathrm{ref}})$. Thus $\mathbb{E}L_d \leq 1 - r$ implies $A_B(d) \geq rA_{B,\mathrm{ref}}$. This common sufficient criterion conservatively ignores replacement readouts' recovery of reference errors.

Fitting and tuning data fix the score models and ordered candidate thresholds. Independent, class-stratified calibration observations test thresholds from conservative to permissive using weighted bounded-loss p-values and fixed-sequence testing [27]. Testing stops at the first failure. Allocating error probability $\delta/5$ per arrival controls population protection across five prespecified sessions for each method, order and partition under independent class-conditional sampling. It does not guarantee arbitrary distribution shifts or every finite test outcome. If no finite threshold passes, the policy returns the reference classifier with zero novel admission. Supplementary Section J provides the bound, assumptions and full selection rule.

A second procedure tests the accuracy requirement directly. For independent calibration pairs, set $Z_i = C_{d,i} - rR_i$, macro weights $w_i = 1/(Bn_{c(i)})$,

and $b = \max_i w_i$. The nonnegative product

$$E_\lambda = \prod_i [1 + \lambda(w_i/b)Z_i], \qquad 0 \le \lambda < 1/r, \tag{9}$$

has expectation at most one under $\sum_i w_i \mathbb{E}Z_i \le 0$. A fixed equal mixture over twelve prespecified values of λ yields a valid p-value $\min(1, 1/E)$ by Markov's inequality. This fixed-sample betting construction [28] credits corrected reference errors and uses the same ordered candidates and five-session error allocation. Supplementary Section N gives the self-contained proof. Calibration images are held out from fitting and tuning; stochastic independence and correspondence to the target class-conditional laws are additional assumptions. The 95% statement applies separately to the five prespecified arrivals of each fixed method/order/partition/policy, rather than jointly to the full experimental grid.

4. Experiments

4.1. Protocol and comparisons

We evaluate CIFAR-100 [29], Oxford-IIIT Pets [30] and ImageNet-R [31], using frozen OpenAI CLIP ViT-L/14 and ViT-B/32 features, five-shot support, ten frequency-stratified class partitions and two five-session arrival orders. Base/novel counts are 40/40, 15/15 and 100/100; each session adds eight, three or 20 classes. Base pools follow long-tailed sampling. ImageNet-R uses task-specific splits after removing byte-identical duplicates. Novelty is relative to task fitting, with CLIP's pre-training exposure retained. Detailed pools, preprocessing and seeds are in Supplementary Sections C, I and K.

Every comparator shares its encoder, partition and support. We compare fixed-gate classifier removal, independently recalibrated removal, dense anchored text–prototype fusion, summed/class-mean Tip-Adapter caches, and raw/fixed-temperature retained fusion. The latter uses a single base-validation scale per branch. HyCal uses the author's heterogeneous cosine/Mahalanobis metrics, count weighting and Gaussian-CDF calibration, transferred to the same eight-prompt anchors and frozen encoders. Its covariance regularization and novel offset are selected by base-only pseudo-expansion. Supplementary Section M documents the pinned implementation, numerical fixtures and transfer settings. All mixtures use base-only selection; FeNeC and earlier three-partition auxiliary studies are retained in Supplementary Sections A–H.

Orders are averaged within a partition. Means and standard deviations describe ten class assignments on fixed images. Paired differences use identical cells; constrained comparisons use jointly feasible sets. Conditional paired 95% bootstrap intervals resample partition means 20,000 times. Average incremental accuracy is the mean macro accuracy over five arrivals. These intervals characterize class-assignment variation, conditional on the fixed data and encoders.

4.2. Novel recognition at a shared base requirement

At the descriptive 95% retention budget, the ImageNet-R L/14 table reaches 84.31% novel accuracy at 86.42% base accuracy. Figure 2 compares novel recognition under shared base requirements across encoders and datasets. A more discriminative base readout can tolerate greater novel competition while meeting the same target, turning retained discrimination into admission capacity.

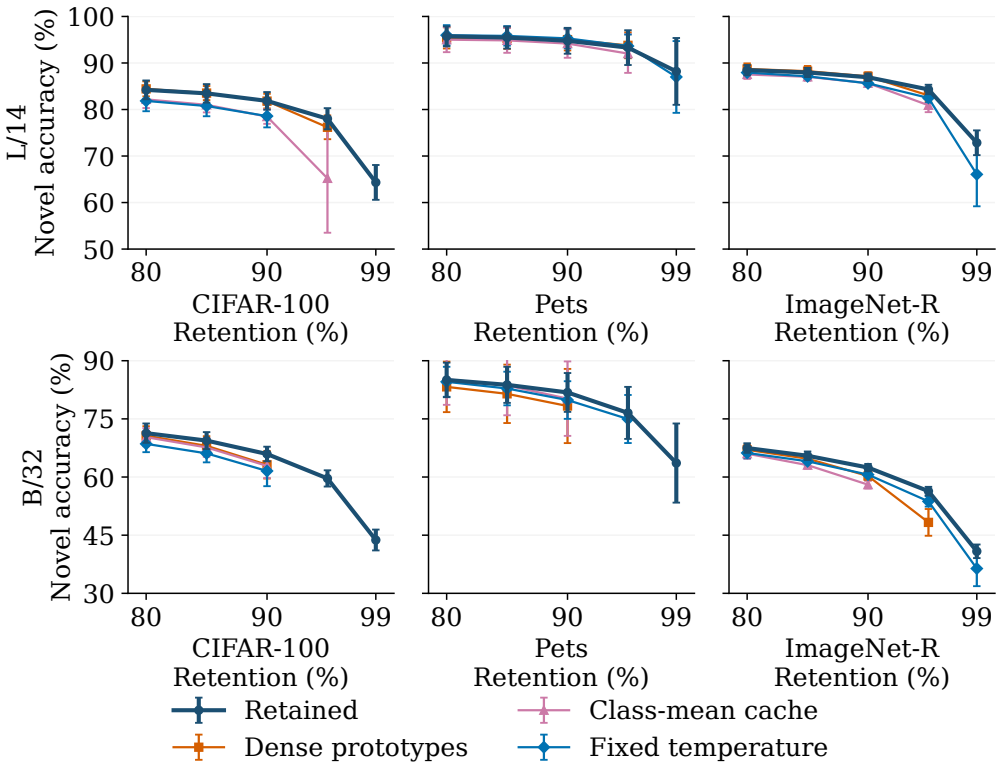


Figure 2: Novel recognition at shared base-retention requirements after the fifth arrival. Each readout must attain the stated fraction of the retained reference's closed-base accuracy within the same partition and encoder. Means and sample standard deviations cover ten partitions after averaging two arrival orders; points require all ten partitions to be feasible. Vertical ranges are 50–100% (L/14) and 30–90% (B/32); error bars stop at panel limits. Test-label threshold sweeps characterize descriptive capacity. Partial feasible sets and exact counts are reported in Supplementary Section I.

4.3. Isolating query-dependent normalization

Table 1 changes centering and scaling independently while fixing the reference banks, learned models, support and original fusion coefficients and offsets. Fixed moments are estimated from base-tuning queries. On ImageNet-R,

query-dependent scaling adds 0.43 [0.17,0.65] and 0.36 [0.15,0.58] points with L/14 and B/32 when query centering is retained. Query centering adds a further 2.03 [1.37,2.68] and 0.88 [0.39,1.55] points at query-dependent scale. These contrasts isolate normalization from score persistence and hyperparameter reselection; the separately tuned fixed-temperature comparator remains in Section I.

Table 1: Controlled normalization factorial. Average incremental macro accuracy (%), mean ± sample standard deviation over ten partitions after averaging two orders. Only branch centering and scaling change; fusion gates and offsets stay fixed. Query/query is the retained table.

Center / scale	CIFAR-100	Pets	ImageNet-R
ViT-L/14			
Query / query	83.30 ± 1.11	91.32 ± 3.44	87.29 ± 0.56
Query / fixed	82.92 ± 1.25	91.20 ± 2.52	86.86 ± 0.45
Fixed / query	80.30 ± 2.31	90.33 ± 3.60	85.25 ± 1.16
Fixed / fixed	78.64 ± 3.00	89.94 ± 3.31	84.16 ± 1.64
ViT-B/32			
Query / query	71.73 ± 1.63	81.14 ± 3.37	67.93 ± 0.33
Query / fixed	71.33 ± 1.56	79.82 ± 3.35	67.57 ± 0.48
Fixed / query	70.35 ± 2.29	78.89 ± 4.56	67.05 ± 0.99
Fixed / fixed	68.99 ± 2.91	77.34 ± 4.65	66.23 ± 1.36

4.4. The classifier's value across arrivals

Table 2 quantifies retained-minus-comparator average incremental accuracy with fixed base-only selected gates and offsets. On ImageNet-R, retaining

the classifier adds 0.71 [0.46, 0.92] points over dense prototypes with L/14 and 0.90 [0.69, 1.11] with B/32. Corresponding gains over class-mean caches are 1.42 [1.15, 1.62] and 1.71 [1.44, 1.98] points. The result persists across both encoders and after calibration is restored to the classifier-free alternative.

Table 2: The retained classifier's value across five arrivals. Cells show paired gains in average incremental accuracy (percentage points), with conditional paired 95% bootstrap intervals below. Positive values favor retention; bold estimates have intervals excluding zero. Two orders are averaged within each of ten partitions. Full final-session differences are in Supplementary Section I.

Comparator	Retained-minus-comparator gain (pp) ↑		
	CIFAR-100	Pets	ImageNet-R
CLIP ViT-L/14			
Recalibrated removal	**+3.00** [+2.23, +3.75]	-0.14 [-1.53, +0.86]	**+2.30** [+1.95, +2.59]
Dense prototypes	+0.15 [-0.46, +0.64]	-0.98 [-2.52, +0.22]	**+0.71** [+0.46, +0.92]
Class-mean cache	**+1.44** [+0.68, +2.10]	-0.06 [-1.76, +1.41]	**+1.42** [+1.15, +1.62]
CLIP ViT-B/32			
Recalibrated removal	**+2.38** [+1.44, +3.34]	-0.44 [-1.64, +0.81]	**+3.07** [+2.82, +3.34]
Dense prototypes	+0.86 [-0.15, +1.76]	-0.39 [-1.85, +1.02]	**+0.90** [+0.69, +1.11]
Class-mean cache	+0.96 [-0.04, +1.84]	-0.74 [-1.62, +0.12]	**+1.71** [+1.44, +1.98]

CIFAR also shows gains over recalibrated removal with both encoders; its dense-prototype intervals span zero. Pets' average incremental intervals span zero for these comparators. The gain over prototype replacement is clearest on ImageNet-R.

The quality diagnosis in Supplementary Section M measures closed-base discrimination before admitting new classes. On ImageNet-R, the retained-minus-dense gap is +0.86 points with L/14 and +2.38 with B/32. Its class-

frequency correlation is small ($\rho = -0.12, +0.06$), and prototype-quality correlations differ in sign ($-0.10, +0.18$). Thus frequency or prototype quality alone does not explain the cross-encoder result. Halving the learned-branch coefficient with all other terms fixed reduces ImageNet-R average incremental accuracy by 2.96 [2.05,3.96] and 3.42 [2.36,4.64] points. The controlled intervention establishes the value of the fitted discrimination in this score frame; class-level associations describe where that value appears. On Pets, the dense-prototype intervals in Table 2 span zero, so the expansion benefit remains dataset-dependent.

The matched HyCal transfer attains 76.36% and 54.19% average incremental accuracy on ImageNet-R with L/14 and B/32. Its final 95% frontier is feasible in two and zero partitions, respectively, requiring both orders to be feasible. Its unrestricted readout and constrained outcomes are reported separately in Section M. These numbers evaluate the controlled transfer to this long-tailed task and shared reference requirement.

4.5. Turning capacity into calibrated admission

An empirical calibration threshold selects the largest offset satisfying r times the reference's calibration accuracy. We separately evaluate whether test accuracy meets r times its test reference. In the original L/14 pools, the retained 95% policy is validation-feasible in all ten partitions but meets the test target in 7/10 CIFAR, 1/10 Pets and 2/10 ImageNet-R partitions. Supplementary Section I reports all methods; Section H varies threshold-label budgets.

For independent risk calibration, the original fitting/validation pool is repartitioned into approximately 40% fitting, 20% tuning and 40% calibration

observations per class. All readouts are refitted and tuned without calibration access (Supplementary Section J). We then freeze the models, support, test archives and 101 tuning-margin threshold candidates. Class-wise calibration caps of 2, 5, 10, 20 and 40 use nested subsets, supplementing the independent pool with unused training images while excluding fitting, tuning, support and test images. Every method receives identical subsets. Three subset seeds are averaged within each partition; the full cap-40 pool is counted once. We use $r = 0.95$ and $\delta = 0.05$. These benchmarks previously informed development; this is a revised benchmark evaluation.

Class coverage determines whether certification is attainable. For macro weights $w_i = 1/(Bn_c)$ and $b = \max_i w_i$, even zero observed loss requires $1/b \geq \log(\delta/5)/\log r = 89.78$ to pass the weighted bound. The original CIFAR and Pets allocations have effective counts 80 and 45–60, respectively. Increasing tail-class coverage makes useful certification possible (Supplementary Section L); total query counts and the derivation are reported there.

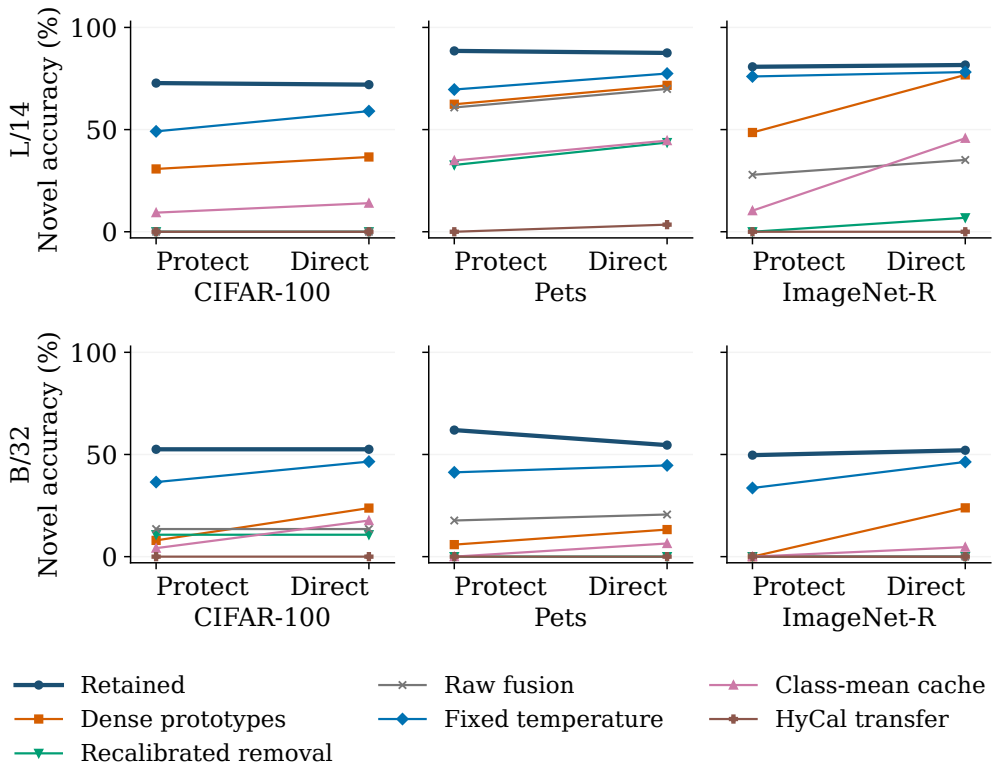


Figure 3: Calibration-rule comparison at identical cap-40 coverage, fifth arrival, first prespecified order. Lines connect the sufficient protection test and the direct paired accuracy test for each readout. Novel means cover all ten partitions, with zero for reference fallback. Each test uses 95% retention and 95% confidence over five prespecified arrivals within a fixed configuration. Admission counts, conditional accuracy and empirical target attainment are reported in Supplementary Section N.

Table 3: Protection-risk-calibrated admission after the fifth arrival in the first prespecified order, with up to 40 images per base class, 95% retention and 95% five-session confidence. Queries are actual base-calibration counts; Finite counts certified expansion and Test met counts attained test targets. Base/Novel are test percentages, mean ± sample standard deviation over ten partitions, including fallback. All six readouts are reported in Supplementary Section L.

Dataset	Calibration	Admission		Test accuracy (%) ↑	
	Queries	Finite	Test met	Base	Novel
CLIP ViT-L/14					
CIFAR-100	1600	10/10	10/10	85.86 ± 1.19	72.76 ± 4.76
Pets	600	10/10	10/10	94.10 ± 2.27	88.51 ± 4.85
ImageNet-R	3636–3778	10/10	10/10	89.02 ± 0.51	80.74 ± 1.83
CLIP ViT-B/32					
CIFAR-100	1600	10/10	10/10	76.74 ± 1.73	52.54 ± 4.34
Pets	600	10/10	10/10	88.10 ± 3.41	61.94 ± 14.07
ImageNet-R	3636–3778	10/10	10/10	72.98 ± 1.21	49.67 ± 2.38

At cap 40, retained tables certify final-session expansion in all ten partitions of each dataset and encoder, with all test targets met (Table 3). L/14 novel accuracy is 72.76% on CIFAR, 88.51% on Pets and 80.74% on ImageNet-R. All 20 arrival paths per dataset/encoder meet every test target. Finite admission holds throughout all CIFAR and ImageNet-R paths and 17/20 L/14 and 16/20 B/32 Pets paths; the remaining Pets paths use an early-session fallback. Full comparisons distinguish policy gains, which include fallback, from accuracy differences on jointly admitted partitions. The common protection criterion

remains conservative for replacement models that recover reference errors.

4.6. Separating recognition from certification

Figure 3 holds calibration images, candidate thresholds and error allocation fixed while changing the statistical criterion. On ImageNet-R L/14, crediting recovered errors raises dense-prototype admission from 9/10 to 10/10 and its fallback-inclusive novel accuracy from 48.58% to 76.76%. Retained tables attain 81.61%, giving a paired gain of 4.85 [2.78,6.93] points on the ten jointly admitted partitions. This remaining difference measures recognition under selected policies; the larger protection-test difference also includes certification effects. The two criteria are complementary: direct calibration admits retained B/32 Pets in 9/10 partitions, compared with 10/10 under protection calibration. One direct-calibrated dense B/32 ImageNet-R partition misses its finite test target. Section N reports both tests, every admission count, conditional accuracy and fallback-inclusive results without selecting a certificate after observing test outcomes.

4.7. Calibration under query distribution shift

A prespecified CIFAR stress test applies Gaussian noise ($\sigma = .08$) or blur (radius 1.25) before encoding, while fitting, support and tuning remain clean. Both encoders evaluate 20 test images per class and the same transformed, separately held-out calibration pool. With the direct test and clean calibration, retained L/14 meets the final shifted test target in 5/10 noise and 1/10 blur partitions. Matching calibration to each shift restores 10/10 for both, with finite admission in every partition and novel accuracy of 31.38% and 12.19%. B/32 matched calibration likewise meets all ten targets, at

8.66% and 10.41% novel accuracy. Complete protection/direct comparisons, including prototype and HyCal readouts, appear in Section O. The stress test uses a previously examined dataset with new query corruptions. It tests the importance of calibration–target correspondence; it supplies neither an untouched application evaluation nor a guarantee for arbitrary shifts.

5. Conclusion

Candidate tables turn retained discrimination into reusable expansion capacity. Fixed references preserve the existing score frame, while an admission threshold exposes the trade-off between original-class protection and novel recognition. Matched-retention and paired sequential comparisons quantify this value. Controlled normalization and learned-weight interventions identify the contribution of the persistent score frame. Recovery-aware calibration separates statistical admission from predictive value, while shifted-query evaluation establishes the role of calibration–target correspondence. Together these results quantify how existing discrimination supports useful, retention-constrained recognition.

CRediT authorship contribution statement

Junyi Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The experiments use public CIFAR-100, Oxford-IIIT Pets and ImageNet-R data and the publicly released CLIP ViT-L/14 and ViT-B/32 checkpoints. Code, source-hashed experimental records, prediction archives, manuscript sources, and table/figure generation scripts are available at `https://github.com/Cheerssom/Reference-Anchored-Candidate-Tables`. Dataset images, pretrained weights, and feature caches are obtained separately using the repository instructions.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used to assist with literature discovery, manuscript drafting and revision, experiment and verification scripts, and LaTeX and table preparation. The author is responsible for the accuracy, interpretation and originality of the submitted work and for reviewing and approving the final manuscript and supporting materials.

References

[1] Z. Liu, Z. Miao, X. Zhan, J. Wang, B. Gong, S. X. Yu, Large-Scale Long-Tailed Recognition in an Open World, arXiv preprint arXiv:1904.05160 (2019).
URL https://arxiv.org/abs/1904.05160

[2] K. Cao, C. Wei, A. Gaidon, N. Arechiga, T. Ma, Learning Imbalanced Datasets with Label-Distribution-Aware Margin Loss, arXiv preprint arXiv:1906.07413 (2019).
URL https://arxiv.org/abs/1906.07413

[3] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, S. Belongie, Class-Balanced Loss Based on Effective Number of Samples, arXiv preprint arXiv:1901.05555 (2019).
URL https://arxiv.org/abs/1901.05555

[4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever, Learning Transferable Visual Models From Natural Language Supervision, in: International Conference on Machine Learning, 2021, pp. 8748–8763.
URL https://proceedings.mlr.press/v139/radford21a.html

[5] K. Zhou, J. Yang, C. C. Loy, Z. Liu, Learning to Prompt for Vision-Language Models, International Journal of Computer Vision 130 (9) (2022) 2337–2348. doi:10.1007/s11263-022-01653-1.
URL https://link.springer.com/article/10.1007/s11263-022-01653-1

[6] J.-X. Shi, T. Wei, Z. Zhou, J.-J. Shao, X.-Y. Han, Y.-F. Li, Long-Tail Learning with Foundation Model: Heavy Fine-Tuning Hurts, in: International Conference on Machine Learning, 2024, pp. 45014–45039.
URL https://proceedings.mlr.press/v235/shi24g.html

[7] J. Snell, K. Swersky, R. S. Zemel, Prototypical Networks for Few-shot Learning, arXiv preprint arXiv:1703.05175 (2017).
URL https://arxiv.org/abs/1703.05175

[8] Y. Wang, W.-L. Chao, K. Q. Weinberger, L. van der Maaten, SimpleShot: Revisiting Nearest-Neighbor Classification for Few-Shot Learning, arXiv preprint arXiv:1911.04623 (2019).
URL https://arxiv.org/abs/1911.04623

[9] R. Zhang, W. Zhang, R. Fang, P. Gao, K. Li, J. Dai, Y. Qiao, H. Li, Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification, arXiv preprint arXiv:2207.09519 (2022).
URL https://arxiv.org/abs/2207.09519

[10] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, C. H. Lampert, iCaRL: Incremental Classifier and Representation Learning, arXiv preprint arXiv:1611.07725 (2016).
URL https://arxiv.org/abs/1611.07725

[11] X. Tao, X. Hong, X. Chang, S. Dong, X. Wei, Y. Gong, Few-Shot Class-Incremental Learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 12183–12192.

URL https://openaccess.thecvf.com/content_CVPR_2020/html/Tao_Few-Shot_Class-Incremental_Learning_CVPR_2020_paper.html

[12] D.-W. Zhou, Z.-W. Cai, H.-J. Ye, D.-C. Zhan, Z. Liu, Revisiting Class-Incremental Learning with Pre-Trained Models: Generalizability and Adaptivity are All You Need, International Journal of Computer Vision (2024).
URL https://github.com/LAMDA-CL/RevisitingCIL

[13] M. D. McDonnell, D. Gong, A. Parvaneh, E. Abbasnejad, A. van den Hengel, RanPAC: Random Projections and Pre-trained Models for Continual Learning, in: Advances in Neural Information Processing Systems, Vol. 36, 2023, pp. 12022–12053. doi:10.52202/075280-0526.
URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/2793dc35e14003dd367684d93d236847-Abstract-Conference.html

[14] X. Xu, S. Niu, Z. Wang, W. Guo, L. Jing, H. Yang, Multi-feature space similarity supplement for few-shot class incremental learning, Knowledge-Based Systems 265 (2023) 110394. doi:10.1016/j.knosys.2023.110394.

[15] R. Chen, X.-Y. Jing, F. Wu, H. Chen, Sharpness-aware gradient guidance for few-shot class-incremental learning, Knowledge-Based Systems 299 (2024) 112030. doi:10.1016/j.knosys.2024.112030.

[16] C. Han, Z. Gu, C. Xu, Z. Cui, Adaptive exploration for few-shot

incremental learning, Knowledge-Based Systems 330 (2025) 114496. doi:10.1016/j.knosys.2025.114496.

[17] R. He, D. Fang, Y. Chen, K. Tong, C. Chen, Y. Wang, L.-p. Chau, H. Zhuang, REAL: Representation enhanced analytic learning for exemplar-free class-incremental learning, Knowledge-Based Systems 332 (2026) 114901. doi:10.1016/j.knosys.2025.114901.

[18] H. Fang, H. Cai, K. Mao, J. Zhao, X. Xiao, Few-shot class-incremental learning based on prompt guidance and multimodal fusion, Knowledge-Based Systems 342 (2026) 115862. doi:10.1016/j.knosys.2026.115862.

[19] L. Li, Z. Wu, Y. Ji, MoTE: Mixture of task-specific experts for pre-trained model-based Class-incremental learning, Knowledge-Based Systems 324 (2025) 113795. doi:10.1016/j.knosys.2025.113795.

[20] M. Dong, Z. Zhang, X. Tan, J. Qiu, Y. Xie, Dynamic expansion orthogonal network for class-incremental learning, Knowledge-Based Systems 341 (2026) 115868. doi:10.1016/j.knosys.2026.115868.

[21] J. Yuan, W. Li, L. Li, L. Zhang, E. Ning, X. Gao, X. Ning, Counterfactual distribution intervention for few-shot class-incremental learning, Knowledge-Based Systems 351 (2026) 116754. doi:10.1016/j.knosys.2026.116754.

[22] Q. Yan, Y. Yang, Y. Dai, X. Zhang, K. Wiltos, M. Woźniak, W. Dong, Y. Zhang, CLIP-guided continual novel class discovery, Knowledge-Based Systems 310 (2025) 112920. doi:10.1016/j.knosys.2024.112920.

[23] K. Książek, H. Jastrzębski, K. Pniaczek, B. Trojan, M. Karp, J. Tabor, FeNeC: Enhancing continual learning via feature clustering with neighbor- or logit-based classification, Knowledge-Based Systems 338 (2026) 115479. doi:10.1016/j.knosys.2026.115479.

[24] Y. Chen, T. Ding, L. Wang, J. Huo, Y. Gao, W. Li, Enhancing few-shot class-incremental learning via training-free bi-level modality calibration, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025, pp. 9881–9890.
URL https://openaccess.thecvf.com/content/CVPR2025/html/Chen_Enhancing_Few-Shot_Class-Incremental_Learning_via_Training-Free_Bi-Level_Modality_Calibration_CVPR_2025_paper.html

[25] E. Lee, M. Kim, J. Kwon, Y. Lee, J. Kim, S. Jang, Y. Kim, Hycal: A training-free prototype calibration method for cross-discipline few-shot class-incremental learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026, pp. 29462–29471.
URL https://arxiv.org/abs/2604.15678

[26] W.-L. Chao, S. Changpinyo, B. Gong, F. Sha, An Empirical Study and Analysis of Generalized Zero-Shot Learning for Object Recognition in the Wild, arXiv preprint arXiv:1605.04253 (2016).
URL https://arxiv.org/abs/1605.04253

[27] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei, Learn then test: Calibrating predictive algorithms to achieve risk control, arXiv preprint arXiv:2110.01052 (2021). doi:10.48550/arXiv.2110.01052.

[28] I. Waudby-Smith, A. Ramdas, Estimating means of bounded random variables by betting, Journal of the Royal Statistical Society Series B: Statistical Methodology 86 (1) (2024) 1–27. doi:10.1093/jrsssb/qkad009.

[29] A. Krizhevsky, Learning multiple layers of features from tiny images, Tech. rep., University of Toronto (2009).
URL https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf

[30] O. M. Parkhi, A. Vedaldi, A. Zisserman, C. V. Jawahar, Cats and dogs, in: IEEE Conference on Computer Vision and Pattern Recognition, 2012, pp. 3498–3505. doi:10.1109/CVPR.2012.6248092.
URL https://www.robots.ox.ac.uk/~vgg/publications/2012/parkhi12a/

[31] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo, D. Song, J. Steinhardt, J. Gilmer, The Many Faces of Robustness: A Critical Analysis of Out-of-Distribution Generalization, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 8340–8349.
URL https://openaccess.thecvf.com/content/ICCV2021/html/Hendrycks_The_Many_Faces_of_Robustness_A_Critical_Analysis_of_Out-of-Distribution_ICCV_2021_paper.html